

Udarata Dance Coaching

Technology and Process Summary

1. What the system is

A Windows desktop coaching application for Kandyan (Udarata) dance. Learners select a step, watch an expert reference video, then practice with a webcam while the system scores **form** and **timing** against a fused expert pose timeline.

2. Technology stack

Layer	Technology
Language / runtime	Python 3
User interface	CustomTkinter (black & gold theme)
Video / camera	OpenCV
Pose estimation	MediaPipe Pose Landmarker (full model)
Numerics	NumPy
Audio playback	Pygame mixer + WAV (imageio-ffmpeg)
Charts / PDF report	Matplotlib, ReportLab
Packaging	PyInstaller (Udarata.exe onedir build)

Core software modules

- **core/pose_extractor.py** — live and offline pose extraction
- **core/angle_calculator.py** — joint angles and deviations
- **core/motion_features.py** — bone directions, velocity, hybrid score
- **core/soft_dtw.py** — Soft-DTW form alignment and lag-based timing
- **core/expert_fusion.py** — multi-expert DTW align, median fuse, variance bands
- **core/dtw_engine.py** — expert JSON loader and matching
- **config.py** — step registry, paths, asset checks
- **preprocess_multi_expert.py** — offline expert rebuild pipeline

3. Dance steps currently supported

Namaskaraya; Pa Saramba 01 / 02 / 03; Goda Saramba 01 / 02 / 03; Goda Saramba Kasthiraama 01 / 02 / 03.

Each step has an expert video (and optional 720p proxy) in **assets/**, a beat WAV beside the video, a fused pose JSON in **data/**, and a **reference_loops** setting (for example Kasthiraama = 2 loops, many other steps = 1, Pa Saramba 01 = 3).

4. Offline process — expert preparation

Before live coaching, expert performances are converted into a fused reference:

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Expert video(s)
  → MediaPipe Pose (VIDEO mode)
  → Joint angles + torso-frame bones
  → DTW-align every expert onto the canonical teaching clip
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- Median fuse + per-joint variance / adaptive tolerance scales
- Save **data/<step>.json**
- Extract WAV audio for synchronized playback

5. Live process — learner session

- **Menu** — select a dance step
- **Preview** — watch expert video with audio
- **Practice** — countdown, then webcam + expert play together
- **Scoring (each frame)** — learner pose $\rightarrow$ angles / bones / velocity; Soft-DTW alignment for Form; timing/lag vs music–video clock; adaptive tolerances from expert variance
- **End** — after N reference loops (or user stop)
- **Report** — Form / Timing summary, joint breakdown, optional PDF export

Hybrid form score (approximate blend): ~55% angles + ~35% bones + ~10% velocity.

6. End-to-end flow

Expert videos → Preprocess / Fusion → Expert JSON + WAV
 Learner webcam ████████████████████ → Live Practice UI ←███ Expert JSON + WAV
 Live Practice UI → Form + Timing scores → Session Report

7. Distribution for testing

A packaged Windows folder can be built with **build_exe.ps1** (PyInstaller). Testers receive **dist/Udarata/** containing **Udarata.exe** plus **assets/**, **data/**, and **models/**. Zip the whole folder and keep that structure when sharing.